
Integrable Systems — Exercise Sheet 2

Two languages: the Poisson bracket and differential forms

Variant: Questions only

Conventions, as in Lecture 2: $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$, so $df/dt = \{f, H\}$; $X_f h = \{h, f\}$, so $X_f = \sum_i (\partial_{p_i} f \partial_{q_i} - \partial_{q_i} f \partial_{p_i})$; $[X, Y]^i = X(Y^i) - Y(X^i)$; and $(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v)$.

Exercise 1 (brackets and angular momentum). On $T^*\mathbb{R}^3$ let $L = q \times p$, that is $L_1 = q_2 p_3 - q_3 p_2$ and cyclically. (i) Compute $\{q_i, p_j\}$, $\{q_i, q_j\}$ and $\{p_i, p_j\}$. (ii) Show that $\{L_i, L_j\} = \sum_k \epsilon_{ijk} L_k$; in particular $\{L_1, L_2\} = L_3$. (iii) Show that each L_i Poisson-commutes with $|q|^2$, $|p|^2$, $q \cdot p$ and $|L|^2$. Compute $\{L_3, q_1\}$ and $\{L_3, q_2\}$, and say what the flow of L_3 does. (iv) Deduce that for every central force, $H = \frac{1}{2}|p|^2 + V(|q|)$ with V smooth, on $q \neq 0$, each L_i is a first integral, and state the Noether reading of this.

Exercise 2 (the engine, in coordinates, for one pair). On $T^*\mathbb{R}$ let $f = q^2 p$ and $g = qp^2$. (i) Compute X_f and X_g . (ii) Compute $[X_f, X_g]$ from the component formula. (iii) Compute $\{f, g\}$ and $X_{\{f, g\}}$, and check the engine $X_{\{f, g\}} = -[X_f, X_g]$. Which sign would the convention $X_f h = \{f, h\}$ produce?

Exercise 3 (first integrals, and Noether). On $T^*\mathbb{R}^2$ take the anisotropic oscillator $H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{2}(q_1^2 + 4q_2^2)$, with frequencies 1 and 2, and let $H_1 = \frac{1}{2}(p_1^2 + q_1^2)$, $H_2 = \frac{1}{2}(p_2^2 + 4q_2^2)$, $L = q_1 p_2 - q_2 p_1$. (i) Which of H_1 , H_2 , L are first integrals? Do H_1 and H_2 Poisson-commute? For the isotropic oscillator $\frac{1}{2}(p_1^2 + p_2^2 + q_1^2 + q_2^2)$, is L a first integral? (ii) Find the flows of H_1 and of L , and check the Noether reading: H is invariant under the one and not under the other. Compute $\frac{d}{ds} H$ at $s = 0$ along the flow of L .

Exercise 4 (covectors). (i) On $\mathbb{R}^2$ let $f = x^2 + y^2$. Write df in the basis dx, dy , and evaluate df at the point $(1, 2)$ on the vectors $v = (3, 1)$ and $w = (-2, 1)$. Interpret the zero in the lecture's picture of a covector as a stack. (ii) In polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, express dx and dy through dr and $d\theta$, and deduce that $x dy - y dx = r^2 d\theta$ and $x dx + y dy = r dr$. (iii) Show that the 1-form $p dq$ on $T^*\mathbb{R}$ is not dF for any smooth function $F(q, p)$.

Exercise 5 (the wedge). (i) Evaluate $(dx \wedge dy)(v, w)$ for $v = (1, 2)$, $w = (3, -1)$, and $(dx \wedge dy)(w, v)$. Which way round do v, w run? (ii) On $\mathbb{R}^2$, compute $(x dx + y dy) \wedge (-y dx + x dy)$. (iii) On $\mathbb{R}^3$, write $(dx + dy) \wedge (dy + dz)$ in the basis $dx \wedge dy$, $dx \wedge dz$, $dy \wedge dz$. (iv) On $\mathbb{R}^4$ with $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$, show that $\omega \wedge \omega = 2 dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2$, and that adding $c dq_1 \wedge dq_2$ to ω (c constant) does not change it. (v) Show that $\alpha \wedge \alpha = 0$ for every 1-form α , and give a 2-form β with $\beta \wedge \beta \neq 0$.

Exercise 6 ★. The Jacobi identity with variable coefficients. On $\mathbb{R}^3$ with coordinates $z = (x, y, z)$, let $J^{\alpha\beta}(z) = -J^{\beta\alpha}(z)$ be smooth, and define $\{u, v\} = \sum_{\alpha, \beta} \partial_\alpha u J^{\alpha\beta} \partial_\beta v$, so that $J^{\alpha\beta} = \{z^\alpha, z^\beta\}$. The lecture's reading proves the Jacobi identity when J is constant. (i) Show that for variable J the terms with second derivatives still cancel, and that the cyclic sum $\mathcal{J}(f, g, h) = \{f, \{g, h\}\} + \text{cyclic}$ satisfies $\mathcal{J}(f, g, h) = \mathcal{J}(x, y, z) \det [\partial_\alpha f, \partial_\alpha g, \partial_\alpha h]$. So Jacobi holds for all functions if and only if it holds for the three coordinates. (ii) Take $\{x_i, x_j\} = \sum_k \epsilon_{ijk} x_k$, the bracket of Exercise 1 (ii) with the L_i as coordinates. Show that it satisfies Jacobi, and that $|x|^2$ Poisson-commutes with every function. (iii) Show that $\{x, y\} = 1$, $\{y, z\} = 0$, $\{z, x\} = x$ violates Jacobi.